INCH-POUND

MIL-R-50090D(MU)
AMENDMENT 2
18 November 1988
SUPERSEDING
AMENDMENT 1
30 August 1982

MILITARY SPECIFICATION

RIOT CONTROL AGENT, CS1

This amendment forms a part of MIL-R-50090D(MU), dated 19 August 1970, and is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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Delete paragraph 4.2.4.2 (including table II and table III) in its entirety and substitute the following:

"4.2.4.2 **Ultimate particle size.**

(a) **Equipment.** Use a Cambridge Instruments Quantimet 920 image analyzing microscope system or an equivalent system, a microscope stage micrometer calibrated in 10-micron and 100-micron intervals, and a four-blade electric stirrer capable of operating at 600 to 2,000 revolutions per minute.

(b) **Program for the Quantimet 920.**

(1) Enter specimen identity
(2) Scanner (No. 1 Vidicon LV = 1.50 SENS = 1.16)
(3) CALL STANDARD
(4) Calibrate microscope (calibration value = 0.3444 microns per pixel)

AMSC N/A FSC 1365

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(5) Detect 2D (darker than 38 PAUSE)
(6) Edit (pause)
(7) Measure feature AREA X.FCP Y.FCP
with limits 0.120 (=AREA 18429
into array FEATURE (of 500 features and 5 parameters)
(8) FEATURE CALC : = 2. * (AREA / PI) 0.50000
(9) Distribution of COUNT CALC (Units MICRONS)
from FEATURE in HIST01 FROM 0.500 to 50.50
in 25 bins (LIN)
(10) Print distribution (HIST01, differential, bar chart, scale = 0.00)
(11) Display HIST01 (LIN) differential PAUSE
(12) Not used
(13) End of program

(c) **Slide preparation.** Weigh 0.150 g of the specimen into each of two 100-milliliter (mL) beakers. Add 20 mL of USP grade mineral oil to the first beaker and mix for 5 ± 1 minutes with an electric stirrer having four blades and a diameter approximately 1/2 inch smaller than the inside diameter of the beaker and operating at a speed of approximately 1,750 revolutions per minute. While continuing to stir the mixture, withdraw a portion of the mixture with an eyedropper, taking it from a point just above the stirrer blades and approximately one-fourth the diameter of the beaker away from the inside wall. Immediately wipe the outside of the eyedropper. Place one drop of the mixture on a clean, dry, 1-inch by 3-inch microscope slide and cover with a No. 1, 3/4-inch diameter cover glass. In the same manner as described above, prepare a second slide from the first beaker. Then following the same procedure with the second beaker, prepare a second pair of slides. Label the four slides to identify the beakers and specimen represented.

(d) **Measurement.** Place the slide on the stage of the microscope and select a field of view. Adjust the focus until a sharp image of the particles is detected. Begin the count. Restart the program and select a second field of view. Continue the selection of fields (approximately 8 to 10 fields) until a total of 1,000 particles are counted."

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Delete paragraph 6.6 in its entirety and substitute the following:

"6.6 **International standardization agreement.** Certain provisions of this specification are the subject of international standardization agreement QSTAG 495. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action

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through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations."

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A096

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